



# Combined effect of a natural flavonoid rutin from *Citrus sinensis* and conventional antibiotic gentamicin on *Pseudomonas aeruginosa* biofilm formation

Murugesan Sathiya Deepika<sup>a</sup>, Ramar Thangam<sup>b, c</sup>, Periasamy Sakthidhasan<sup>d</sup>,  
Sridhar Arun<sup>a</sup>, Srinivasan Sivasubramanian<sup>c</sup>, Ramasamy Thirumurugan<sup>a, \*</sup>

<sup>a</sup> Department of Animal Science, School of Life Sciences, Bharathidasan University, Tiruchirappalli 620 024, Tamil Nadu, India

<sup>b</sup> CSIR-Central Leather Research Institute, Chennai 600 020, Tamil Nadu, India

<sup>c</sup> King Institute of Preventive Medicine & Research, Chennai 600 032, Tamil Nadu, India

<sup>d</sup> Department of Plant Science, School of Life Sciences, Bharathidasan University, Tiruchirappalli 620 024, Tamil Nadu, India

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## ABSTRACT

Extensive and irrational use of antibiotics to treat biofilm associated infections has led to the emergence of multiple drug resistant (MDR) strains. Rutin, a vitamin P rich flavonoid group of phytochemical with antimicrobial property, has been studied for its anti-biofilm potential against *Pseudomonas aeruginosa*, which is a model biofilm forming pathogenic bacterium, in combination with conventional antibiotic gentamicin. For this study, rutin was isolated from the peels of *Citrus sinensis* by column chromatography and HPLC and characterized by UV-vis and FT-IR spectral analyses. Results showed that rutin exhibited minimum inhibitory concentration (MIC) at 800 µg/ml against the bacterium. Its anti-biofilm potential was examined by crystal violet staining, protein analysis, microscopy studies and quantification of exopolysaccharide (EPS) using several sub-MIC doses. The inhibition of biofilm formation was synergistically enhanced when the bacterium was treated with rutin (200 µg/ml) along with an antibiotic gentamicin (2.5 µg/ml). EPS production was responsible for adherence and maturation of biofilms, which was also significantly reduced in the tested strain. To the best of our knowledge, this is the first study describing the anti-biofilm potential of citrus flavonoid rutin as well as its synergistic effects when combined with gentamicin against *P. aeruginosa*. Findings of the study substantiate that rutin can be potentially used as an adhesion inhibitor for the treatment of *P. aeruginosa* infections.

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## 1. Introduction

Though the peels of *Citrus sinensis* are regarded as waste, they have several nutraceutical properties due to the presence of several flavones and flavanones that are uncommon in other plants (Ortuño et al., 2006). Rutin is one of the major flavonoid compound of *C. sinensis* peel, and it is also known as quercetin-3-O-rutinoside, rutoside, sophorin and vitamin P (Ganeshpurkar & Saluja, 2016). Rutin is also an essential nutritional constituent of plant based food substances and has evinced number of pharmacological activities, including cytoprotective (Moridani, Pourahmad, Bui, Siraki, &

O'Brien, 2003), vasoprotective, hypotensive (da Rocha Lapa et al., 2011), cardioprotective (Ali, Mudagal, & Goli, 2009), neuro-protective (Almaas, Saugstad, Pleasure, & Rootwelt, 1998), antioxidant (Kurisawa, Chung, Uyama, & Kobayashi, 2003; Lue et al., 2010), anti-inflammatory (Lee et al., 2015), antiallergic (de Oliveira, Fernandes, & Vieira, 2006), antiviral (Zandi et al., 2014), antimicrobial (Aruna, Geetha, & Suguna, 2009), antiplatelet (Sheu, Hsiao, Chou, Shen, & Chou, 2004), antimutagenic (Kaur, Arora, & Thukral, 2015), anticarcinogenic (Martínez Conesa et al., 2005; Alonso-Castro, Domínguez, & García-Carrancá, 2013), antidiabetic (Jadhav & Puchchakayala, 2012), anti-osteoclastogenic (Kyung, Lee, Shin, & Choi, 2008), wound healing (Tran, Joung, Lih, & Park, 2011) and potent scavenger of superoxide radicals (Xie, Yang, Tang, Chen, & Ren, 2015) etc. Rutin has been isolated from different plants using a variety of separation techniques (Wang, Wu, Zu, & Fu, 2008; Fathiazad, Delazar, Amiri, & Sarker, 2010; Jiang et al., 2007; Shukla,

\* Corresponding author.

E-mail addresses: [ramthiru72@bdu.ac.in](mailto:ramthiru72@bdu.ac.in), [ramthiru72@gmail.com](mailto:ramthiru72@gmail.com) (R. Thirumurugan).

Gopalkrishna, & Shukla, 2012; Xie et al., 2015). Nevertheless, the necessity for identifying a better commercial source for rutin is still unfulfilled. Besides, flavonoids are reported to exhibit beneficial antioxidant properties to the host systems without toxicity but they serve as pro-oxidants against bacterial pathogens and induce oxidative stress through generating reactive oxygen species (ROS) to cause death of the bacteria (Fathima & Rao, 2016). Hence, this study attempts to explore the mechanism of antibacterial as well as antibiofilm activities of rutin isolated from *C. sinensis* by means of its potential to generate ROS via inducing oxidative stress in bacterial pathogens.

Infectious disease remains the second most common cause of death worldwide and the majority of these deaths are caused by bacterial-related infections (Jones et al., 2008). Numerous studies have emerged on the failure of antimicrobial therapy due to the development of bacterial resistance against multiple antibiotics (Jones et al., 2008; Simoes, 2011). In cases concerning chronic infections, the failure to achieve complete bacterial eradication with antibiotics is largely due to the switch of bacterial growth mode from free-swimming planktonic cells into sessile community structured biofilms (Loo et al., 2015). Biofilms are complex microbial communities adhered to surfaces, which have been found to be involved in 80% of all microbial infections in the body (Das et al., 2016). Biofilms potentially afflict immune system due to the altered expression of polysaccharide matrix and the phagocytosis-resistant feature of biofilm-colonies (Vikram, Jayaprakasha, Jesudhasan, Pillai, & Patil, 2010). Among all biofilm producing microbial pathogens, *Pseudomonas aeruginosa* causes the most challenging infection due to its biofilm forming ability on human host that results in kidney infections, urinary tract infections, cystic fibrosis, etc (Hall-Stoodley, Costerton, & Stoodley, 2004). *P. aeruginosa* has been most intensively studied by researchers for the reason that this bacterium is an opportunistic human pathogen and is responsible for the death of thousands of cystic fibrosis sufferers and immune-compromised patients (Annapoorani, Umamageswaran, Parameswari, Pandian, & Ravi, 2012). The current conventional antibiotic therapies against *P. aeruginosa* infections lack efficacy, which is mainly due to the pathogen's capability to produce exopolysaccharide (EPS) with varying diversity and its associated biofilm formation for conferring increased resistance to antibiotics and evading host immune responses (Sultana, Call, & Beyenal, 2016). Therefore, there is an urgent need for revamping the treatment measures to counter *P. aeruginosa* pathogenesis as well as to control the emergence of high virulent strains through irrational and extensive use of antibiotics. Previous studies have focused on the anti-biofilm potential of various phyto compounds especially flavonoids including a study that reports biofilm inhibitory activity of rutin against drug resistant biofilm producing microbial pathogens (Al-Shabib et al., 2017; Singh, Gupta, & Pandey, 2016; Wang et al., 2017). So far, no literature is available on demonstrating the anti-biofilm activity of rutin against *P. aeruginosa*. In the current study, we isolated the rutin from peels of *Citrus sinensis* using chromatographic techniques and explored its potential as a broad spectrum biofilm control agent against multidrug-resistant (MDR) *Pseudomonas aeruginosa* strain.

## 2. Materials and methods

### 2.1. General chemicals, reagents, and test microorganisms

Chemicals and solvents used for the extraction and purification of rutin were of laboratory reagent grade. Chromatographic isolation of the study compound was done using silica gel #60–120 (Merck Specialties Private Limited, Mumbai, India). Solvents such as acetonitrile and methanol used for HPLC analysis were of HPLC

grade (Merck Specialties Private Limited, Mumbai, India). Rutin standard was obtained from Loba Chemie, Mumbai, India. 24 and 96-well polystyrene plates, glass slides ( $1 \times 1$  cm), Gentamicin sulfate, Luria Bertani (LB) media, LB agar, 2',7'-Dichlorofluorescein diacetate (DCFDA), ascorbic acid (AA), Acridine orange and dimethyl sulfoxide (DMSO for rutin preparation at the concentration of  $1 \text{ mg ml}^{-1}$ ) were obtained from Sigma-Aldrich. Ethanol, glycerol, sodium hydroxide (NaOH), Crystal Violet (CV), ethylenediamine-tetra acetic acid (EDTA), phenol and sulfuric acid ( $\text{H}_2\text{SO}_4$ ) were procured from HiMedia Laboratories Private Limited, Mumbai. The test microorganism *P. aeruginosa* (MTCC 2488) was procured from MTCC (Microbial Type Culture Collection), Chandigarh, India.

### 2.2. Plant materials

Orange peels (*C. sinensis*) were collected and air dried in the shade for 7 days at room temperature. The dried peels were cut into small pieces and ground to a coarse powder by the electric grinder. The powder was sieved by 1 mm pore size sieve and stored in airtight plastic containers at room temperature until further use.

### 2.3. Preparation of extracts

100 g of orange peel powder was used to extract rutin by using different solvents with an increasing order of polarity viz. hexane, chloroform, ethyl acetate and methanol for 6 h in a Soxhlet apparatus. The extraction was done for 6 h and all the extracts were filtered through a Whatman No. 1 filter paper to remove residues. Each of the purified extract was concentrated under vacuum on a rotary evaporator (Buchi, Flawil, Switzerland) with a temperature-controlled bath. To confirm the presence of flavonoids, a small quantity of magnesium ribbon and few drops of concentrated HCl were added to all the different extracts and heated in boiling water (De, Dey, & Ghosh, 2010). Formation of red color indicates the presence of flavonoids in ethyl acetate extract; further, the extract was subjected to chromatographic studies for the separation of flavonoid (rutin) compound.

### 2.4. Isolation of flavonoid compound

The ethyl acetate extract of *C. sinensis* (1.6 g) was subjected to column chromatography (CC) using silica gel (#60–120) as the stationary phase. The charged column was then eluted with n-hexane: petroleum ether (50:50) (E1), chloroform (E2), chloroform: ethyl acetate (50:50) (E3), ethyl acetate (E4), ethyl acetate: methanol (50:50) (E5) and the fractions were concentrated in a rotary evaporator at  $45^\circ\text{C}$ . All the five fractions (E1, E2, E3, E4 and E5) were subjected to analytical HPLC analysis using methanol: acetonitrile, 70:30 (v/v) as the mobile phase to isolate pure compound (HPLC system with the Model LC-2010 CHT with pump model 2 LC-10 ADvp, Column model CTO-10 A (C) vp and Autosampler model SIL-10 ADvp (Shimadzu, Japan)). Fraction E3, eluted by using 50:50 ratio (v/v) of chloroform and ethyl acetate, was filtered through a syringe filter (0.22  $\mu\text{m}$  pore size) for preparative HPLC studies. The filtrate was loaded onto a C-18 column (Sunfire 5  $\mu\text{m}$ , 150 mm  $\times$  4.6 mm; Waters) with the injection volume of 20  $\mu\text{l}$  at a flow rate of 1.0 ml/min to resolve the compound rutin (6 mg) and the eluates were monitored at 356 nm. Preparative HPLC was employed for optimum separation of compound in the ethyl acetate extract with the following conditions: extract: 20  $\mu\text{l}$ , 10 mg/ml; C-18 HPLC column (4.6  $\times$  250 mm; 5  $\mu\text{m}$ ); mobile phase: methanol/acetonitrile (70/30, v/v) gradient over 20 min; flow rate: 1.2 ml/min; and detection: 356 nm. The retention time of standard was used to discern the compound in the extract, which was

further confirmed by characterization studies.

## 2.5. Characterization studies

### 2.5.1. UV-Vis spectrometric analysis

For UV-Vis spectrometric analysis, the isolated compound was dissolved in methanol and subjected to single beam spectrophotometer (Biodrop, Cambridge, UK (UV-1100)) in the wavelength range between 200 nm and 700 nm. Methanol was used as a reference solvent.

### 2.5.2. FT-IR analysis

The FT-IR spectra were carried out using 400 FT-IR Spectrometer (Perkin Elmer, USA). The functional groups of the isolated compound were detected by KBr pellet technique with a scan rate of 5 per min at the resolution of  $4\text{ cm}^{-1}$  in the wave region  $450\text{--}4000\text{ cm}^{-1}$ .

## 2.6. Bacterial strain and culture conditions

The bacterium *P. aeruginosa* MTCC 2488, reported to show resistance to all major antibiotics used in the treatment of nosocomial infections (Mesaros et al., 2007), was used in the current study. The bacteria were cultured in Luria Bertani (LB) medium. Prior to that the bacterial strain was streaked from  $-80^\circ\text{C}$  glycerol stock onto LB agar plate and single colony was inoculated into LB media and incubated at  $37^\circ\text{C}$  for 24 h. For all subsequent experiments,  $10^6$  CFU/ml bacterial cell suspensions were used.

## 2.7. MIC determination

MIC of rutin and gentamicin against biofilm forming MDR strain *P. aeruginosa* were determined using the standard micro broth dilution method according to Clinical and Laboratory Standards Institute standards (Wayne, 2007), and all further analyses were done at sub-MIC levels. Experiments were performed in triplicates.

## 2.8. Synergy studies

Rutin was mixed with gentamicin to test their synergistic effects against the bacterium. This study was performed by a checker board titration method using 96-well polypropylene microtitre plate. The fractional inhibitory concentration (FIC) index was equal to (MIC of gentamicin in combination/MIC of gentamicin alone) + (MIC of rutin in combination/MIC of rutin alone). The definitions of synergism, indifference and antagonism were interpreted as  $\text{FICI} \leq 0.5$ ,  $>0.5$  and  $\leq 4.0$ , and  $>4.0$ , respectively (Orhan, Bayram, Zer, & Balci, 2005).

Initially, different sub MICs of gentamicin (MIC:  $10\text{ }\mu\text{g/ml}$ ;  $5\text{ }\mu\text{g/ml}$  ( $1/2$  MIC),  $2.5\text{ }\mu\text{g/ml}$  ( $1/4$  MIC) and  $1.25\text{ }\mu\text{g/ml}$  ( $1/8$  MIC)) were examined to determine the minimum concentration that could be used synergistically with rutin. To study the effect of rutin and gentamicin combination against the test pathogen, the sub-MIC of  $2.5\text{ }\mu\text{g/ml}$  ( $1/4$  MIC) was selected (Table 1).

## 2.9. Growth curve analysis

The growth curve of *P. aeruginosa*, in the presence and absence of sub-MIC doses of rutin ( $50\text{ }\mu\text{g/ml}$  ( $1/16$  MIC),  $100\text{ }\mu\text{g/ml}$  ( $1/8$  MIC),  $200\text{ }\mu\text{g/ml}$  ( $1/4$  MIC),  $400\text{ }\mu\text{g/ml}$  ( $1/2$  MIC), and sub-MIC dose of gentamicin ( $2.5\text{ }\mu\text{g/ml}$  ( $1/4$  MIC)) with respect to untreated control was used for analyzing antimicrobial potential according to the method described by Dheilly et al. (2010) with slight modifications (Table 1). Briefly,  $10\text{ }\mu\text{l}$  of test inoculums were taken from bacterial suspensions ( $10^6$  CFU/mL suspensions) and added to the wells of 96-well microtitre plate containing sub-MIC dose of test compounds in  $250\text{ }\mu\text{l}$  of LB. The plates were incubated at  $37^\circ\text{C}$  and the growth rate was measured using UV-Vis spectrophotometer at  $600\text{ nm}$  at 3 h intervals up to 24 h. The assays were performed in triplicates.

## 2.10. Quantification of EPS production

EPS content was measured by previously described method (Tribedi & Sil, 2014). Bacterial strain was grown in the presence and absence of rutin (in sub-MICs) along with antibiotic (Table 1), and centrifuged at  $3500\text{ g}$  for 20 min at  $4^\circ\text{C}$ . The resulting supernatants were collected and the pellets were treated with  $2\text{ ml}$  of  $10\text{ mM}$  EDTA. Then, it was vortexed for 15 min and re-centrifuged again to extract cell-bound EPS. The supernatant was collected and mixed with the previous supernatant. The pooled supernatants were mixed with  $2.2$  volume of cold absolute ethanol and incubated at  $-20^\circ\text{C}$  for 1 h and centrifuged at  $3500\text{ g}$  for 20 min at  $4^\circ\text{C}$ . The pellet containing EPS was dissolved in sterile water and quantified by the method of Dubois (1956). Briefly,  $0.05\text{ ml}$  of  $80\%$  phenol was added with  $2\text{ ml}$  of EPS in a tube followed by the addition of  $5\text{ ml}$  of concentrated  $\text{H}_2\text{SO}_4$ . The tubes were allowed to stand for 10 min, vortexed for 2 min, placed for 10–20 min in a water bath at  $25^\circ\text{C}$  and recorded the absorbance at  $490\text{ nm}$ .

## 2.11. Biofilm formation assay

Sub-MICs ( $50\text{ }\mu\text{g/ml}$  ( $1/16$  MIC),  $100\text{ }\mu\text{g/ml}$  ( $1/8$  MIC),  $200\text{ }\mu\text{g/ml}$  ( $1/4$  MIC),  $400\text{ }\mu\text{g/ml}$  ( $1/2$  MIC)) of the rutin and gentamicin ( $2.5\text{ }\mu\text{g/ml}$  ( $1/4$  MIC)) were used for evaluation of anti-biofilm formation as described by (Cassat, Lee, & Smeltzer, 2007). A loopful of overnight grown culture was transferred to LB broth and incubated overnight at  $37^\circ\text{C}$   $50\text{ }\mu\text{l}$  of culture was transferred to LB and adjusted to  $\text{OD}_{600}$  of  $0.1$  ( $10^6$  CFU/ml suspensions). Aliquots of the suspension ( $100\text{ }\mu\text{l}$ ) were added into 96-well flat-bottomed polystyrene plate and incubated at  $37^\circ\text{C}$  for 48 h. Then, the plates were gently washed three times with  $1\text{X}$  phosphate buffered saline (PBS;  $\text{pH } 7.4$ ). The biofilm formation was quantified using crystal violet assay and the absorbance was measured at  $590\text{ nm}$ . Tests were carried out in triplicates and the mean biofilm of absorbance value was used to assess the biofilm forming capacity of the strain. To find out the percentage of biofilm inhibition in all treated wells with respect to untreated control, the following formula was used:

**Table 1**  
Identified MIC values of rutin (R) and gentamicin (Gen). Selection of sub-MIC doses and combinations for this study.

| Compounds  | MIC ( $\mu\text{g/ml}$ ) | Selected sub-MIC ( $\mu\text{g/ml}$ ) | Concentrations selected for this study                 |                                                         |                                                         |                                                         |
|------------|--------------------------|---------------------------------------|--------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------------|
| Rutin      | 800                      | 50, 100, 200, 400                     | R50 $\mu\text{g/ml}$                                   | R100 $\mu\text{g/ml}$                                   | R200 $\mu\text{g/ml}$                                   | R400 $\mu\text{g/ml}$                                   |
| Gentamicin | 10                       | 2.5                                   | R50 $\mu\text{g/ml}$ + Gen $2.5\text{ }\mu\text{g/ml}$ | R100 $\mu\text{g/ml}$ + Gen $2.5\text{ }\mu\text{g/ml}$ | R200 $\mu\text{g/ml}$ + Gen $2.5\text{ }\mu\text{g/ml}$ | R400 $\mu\text{g/ml}$ + Gen $2.5\text{ }\mu\text{g/ml}$ |

$$\text{Biofilm inhibition (\%)} = \frac{(\text{OD of untreated control}) - (\text{OD of treated sample})}{\text{OD of untreated control}} \times 100$$

### 2.12. Estimation of total protein concentration in biofilm

Concentration of extractable protein in biofilms was determined for the bacterium (treated and untreated). The microbial population density in biofilm is directly proportional to the amount of extractable protein concentration (Tribedi & Sil, 2014). Hence, to estimate the total extractable protein from tube surface, the bacterial strain was inoculated at 37 °C for 48 h with or without sub-MIC doses of rutin alone and in combination with antibiotic (Table 1). After incubation, the planktonic cells were removed; adhered cells in biofilm were washed gently with sterile PBS (pH 7.4) and boiled for 30 min in 5 ml of 0.5 N NaOH to extract protein. After that, the suspension was centrifuged at 10,000 rpm for 5 min and the resulting clear supernatant was collected. From the supernatant, protein concentration was measured (Lowry, Rosebrough, Farr, & Randall, 1951).

### 2.13. In situ visualization of biofilm formation through microscopic techniques

#### 2.13.1. Morphological analysis of biofilm development using light microscopy

The qualitative effect of rutin and their combination with gentamicin on biofilm formation was assessed using light microscope. *P. aeruginosa* was allowed to form biofilm on the glass slides (1 × 1 cm) in 24-well polystyrene plates supplemented with sub-MIC doses of rutin and antibiotic. The cultures were then incubated for 24 h at 37 °C. The slides containing biofilm were stained with 0.4% Crystal Violet for 1 min, washed with distilled water to remove excess stain and allowed to dry. The glass slides were then viewed at 200× magnification Carl-Zeiss (Jena, Germany) Axioscop-2 Plus research microscope and documented with an attached digital camera (Sony DXC-151AP CCD camera, Tokyo, Japan).

#### 2.13.2. Confocal laser scanning microscopic analysis

Biofilms were grown on glass slides (1 × 1 cm) in 24-well polystyrene plates supplemented with sub-MIC doses of rutin and antibiotic. The cultures were then incubated for 24 h at 37 °C and gently washed three times with 0.1 M PBS (pH 7.4) to remove non-adherent *P. aeruginosa* cells and stained with 0.1% acridine orange. The excess stain was washed out and the plates were air dried before examination. Glass slides with cells grown in rutin-gentamicin free medium were utilized as control. Images of the stained slides were visualized under CLSM (Model LSM 710, Carl Zeiss, Germany) and processed with Zeiss LSM Image Examiner (Version 4.2.0.121), equipped with an excitation filter 515–560 nm. CLSM images were obtained from triplicates of untreated control and treated biofilms (the experiment was repeated at least thrice) and the Z-stack analysis (surface topography and three-dimensional architecture) was performed with the Zen 2009 software (Carl Zeiss, Germany). Maximum thickness of the biofilms formed on glass surface was evaluated using Z-stack imaging (Scale bars = 50 μm).

### 2.14. Effect of rutin-gentamicin on endogenous ROS production

The ability of rutin-gentamicin to induce endogenous radical

oxygen species (ROS) in *P. aeruginosa* was evaluated using an oxidation-sensitive fluorescent probe 2',7'-dichlorofluorescein diacetate (DCFDA). Briefly, the planktonic cells after treatment with rutin for 24 h at their sub-MIC doses were washed twice with sterile PBS, and incubated with DCFDA (20 μM) for 1 h at 37 °C. Fluorescence intensity was measured using a Synergy HTX multi-mode microplate reader (BioTek Instruments, Inc., Winooski, VT, U.S.A) at an excitation wavelength of OD<sub>492 nm</sub> and an emission wavelength of OD<sub>525 nm</sub> (Bink et al., 2011). Further, the impact of ROS with respect to the anti-biofilm activity of rutin and gentamicin was studied in the presence and absence ROS-inhibitor Ascorbic acid (AA) (50 μg/ml). The concentration of ROS inhibitor scavenged free radicals up to ≥90% (DPPH assay at OD<sub>517 nm</sub>) without affecting the cell density of the test pathogen at OD<sub>600 nm</sub>.

### 2.15. Statistical analysis

Data are presented as mean ± SD. Experiments were performed in triplicates and the statistical analyses were executed by one-way analysis of variance (ANOVA). Significant differences between groups were discerned at  $p \leq 0.05$ . Statistic software Graph-pad Prism 5.0 (Graph-pad Software Inc., San Diego, CA) was used for all the graphical and statistical evaluations.

## 3. Results and discussion

### 3.1. Isolation of rutin

The structural elucidation of flavonoid rutin was performed in the present study using appropriate characterization methods, and the total yield of the isolated rutin was found to be 6 mg/100 g from the peel of *C. sinensis*. Though different extracts were prepared using soxhlet apparatus, the ethyl acetate extract yielded maximum quantity of flavonoids as indicated by appearance of intense red color. Further, the ethyl acetate extract was fractionated with solvents such as hexane, petroleum ether, chloroform, ethyl acetate and methanol using column chromatography to elute the target compound. While subjecting the ethyl acetate fraction (E3) for HPLC analysis, the compound rutin was separated with the retention time (RT) of 2.659 at the detection wavelength of 356 nm and the RT was compared with the RT of rutin standard (Fig. 1, and inset Fig. 1). Vachirapatama & Chamnankid (2012) reported that the rutin isolated from apple juices had the retention time of 2.92 min, which is closely correlated with RT of rutin of our current study. Isolated rutin appeared as light yellow solid upon drying and showed a melting point at a range of 180–190 °C which was similar to the report of Ashok and Saini (2013). The characterization of rutin was achieved through UV-Vis and FT-IR spectrometric analyses (Figs. 2 and 3). The UV-Vis spectra of isolated compound showed the absorption at about 360 nm, which was similar to the standard rutin molecule. The absorption range indicated the presence of flavonol structure as reported by Pawar & Salunkhe (2013) and Shukla et al. (2012). FT-IR spectra showed the peaks at (KBr)  $V_{\max} \text{ cm}^{-1}$ : 3422 (O-H stretching-H bonded), 2936  $\text{cm}^{-1}$  (N-H,  $\text{CH}_2$ -stretching), 1600, 1656  $\text{cm}^{-1}$  (C=C, C=O, C=N stretching), 1456  $\text{cm}^{-1}$  (C=O groups), 1362  $\text{cm}^{-1}$  (C-OH vibrations), 1295  $\text{cm}^{-1}$  (C=N, C-O-C stretching), and 1122  $\text{cm}^{-1}$  (C-O stretching) (Fig. 3). Based on the functional groups and stretches, the FT-IR spectrum

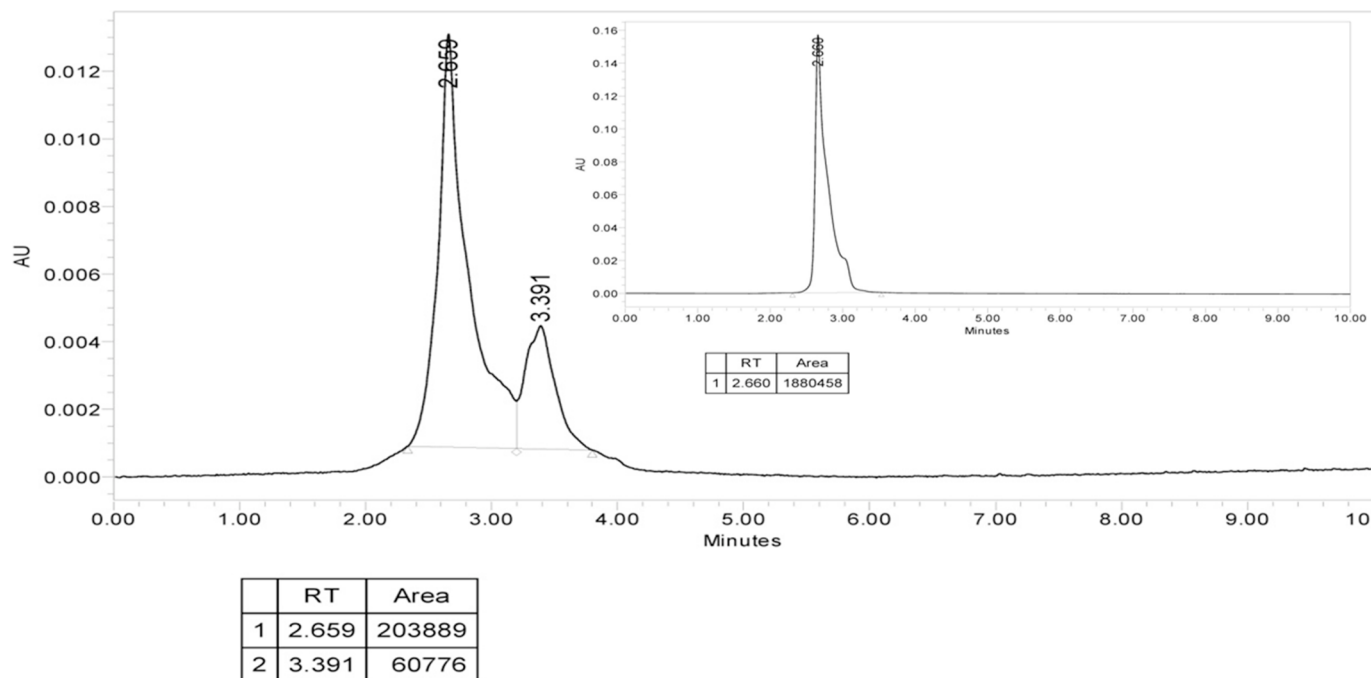


Fig. 1. HPLC profile of isolated compound rutin as compared with standard rutin (inset Fig.).

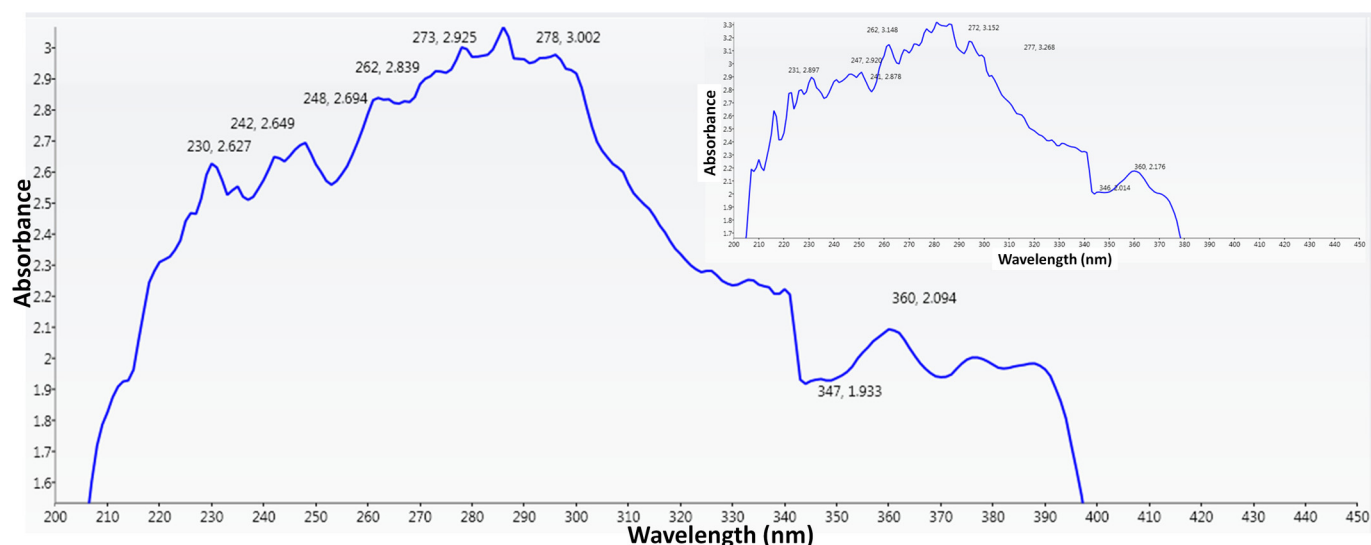


Fig. 2. UV-Vis spectrum of isolated compound rutin as compared with standard rutin (inset Fig.).

results proved the presence of flavonoid, and these findings were in accordance to previously published report (Selvaraj, Chowdhury, & Bhattacharjee, 2013).

### 3.2. Evaluation of anti-biofilm potential

Biofilm is a mass community of microbial cells that attach together through secreted polymers and grow on living or nonliving surfaces. Biofilm related bacterial infections are often hard to treat due to their MDR property. Therefore, it is essential to find new or efficient molecules against bacterial biofilm formation (Kumar, Chhibber, & Harjai, 2013). Moreover, inhibition of biofilm is the first line of defense mechanism which controls surface adhered bacterial population growth and survival. Since rutin was reported

to have antimicrobial activity (Amin, Khurram, Khattak, & Khan, 2015; Askun, Tumen, Satil, & Ates, 2009; Cushnie & Lamb, 2005; Orhan, Özçelik, Özgen, & Ergun, 2010), we have evaluated the anti-biofilm activity of rutin against model biofilm forming pathogenic organism *P. aeruginosa* in the present study. We have also examined the effect of sub-MIC concentrations of conventional antibiotic gentamicin on anti-biofilm activity in combination with rutin against *P. aeruginosa*.

#### 3.2.1. Antimicrobial activity of rutin and gentamicin

Antimicrobial activity of rutin and gentamicin against *P. aeruginosa* were assessed by broth micro-dilution assay to find out MIC values. The MIC value of rutin against *P. aeruginosa* was found to be 800 µg/ml, whereas gentamicin shows MIC at 10 µg/ml

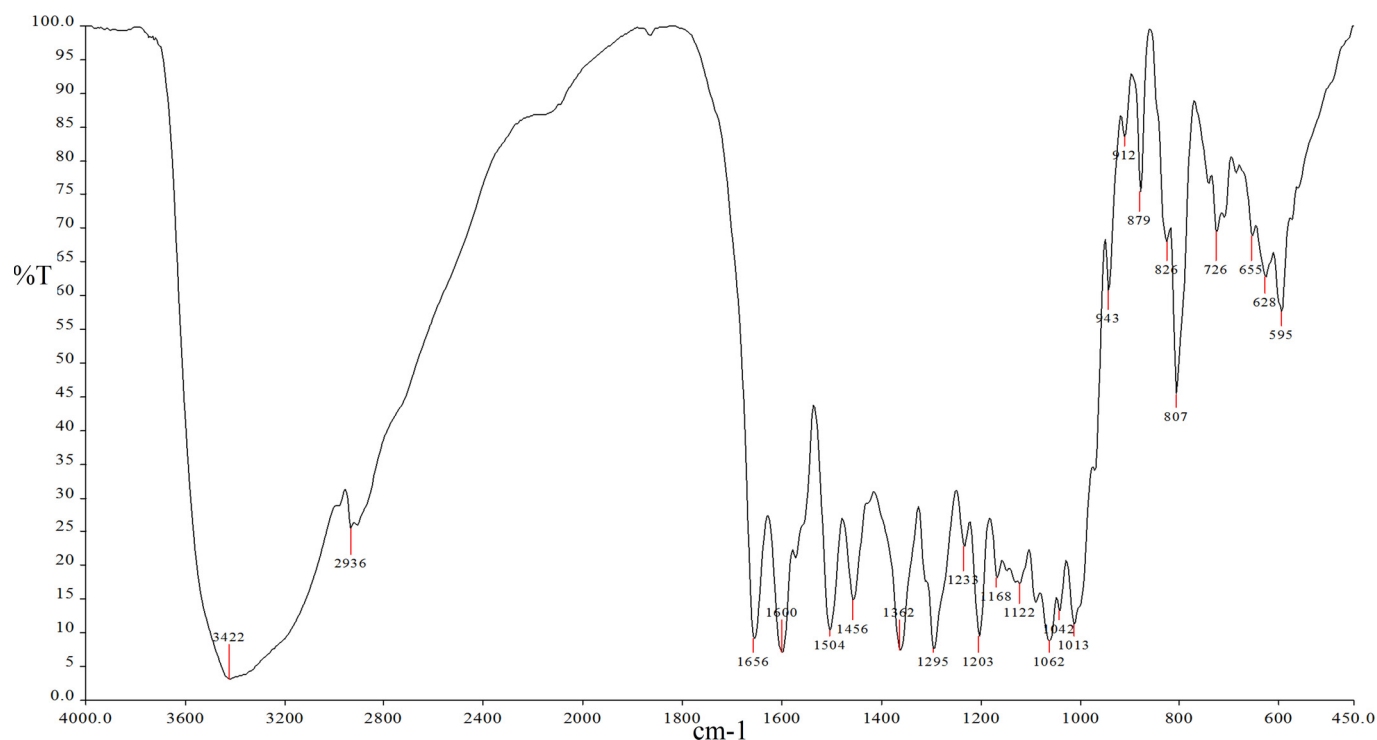


Fig. 3. FT-IR spectrum of isolated compound rutin.

and these results suggest that rutin has moderate antimicrobial activity against *P. aeruginosa* (Table 1).

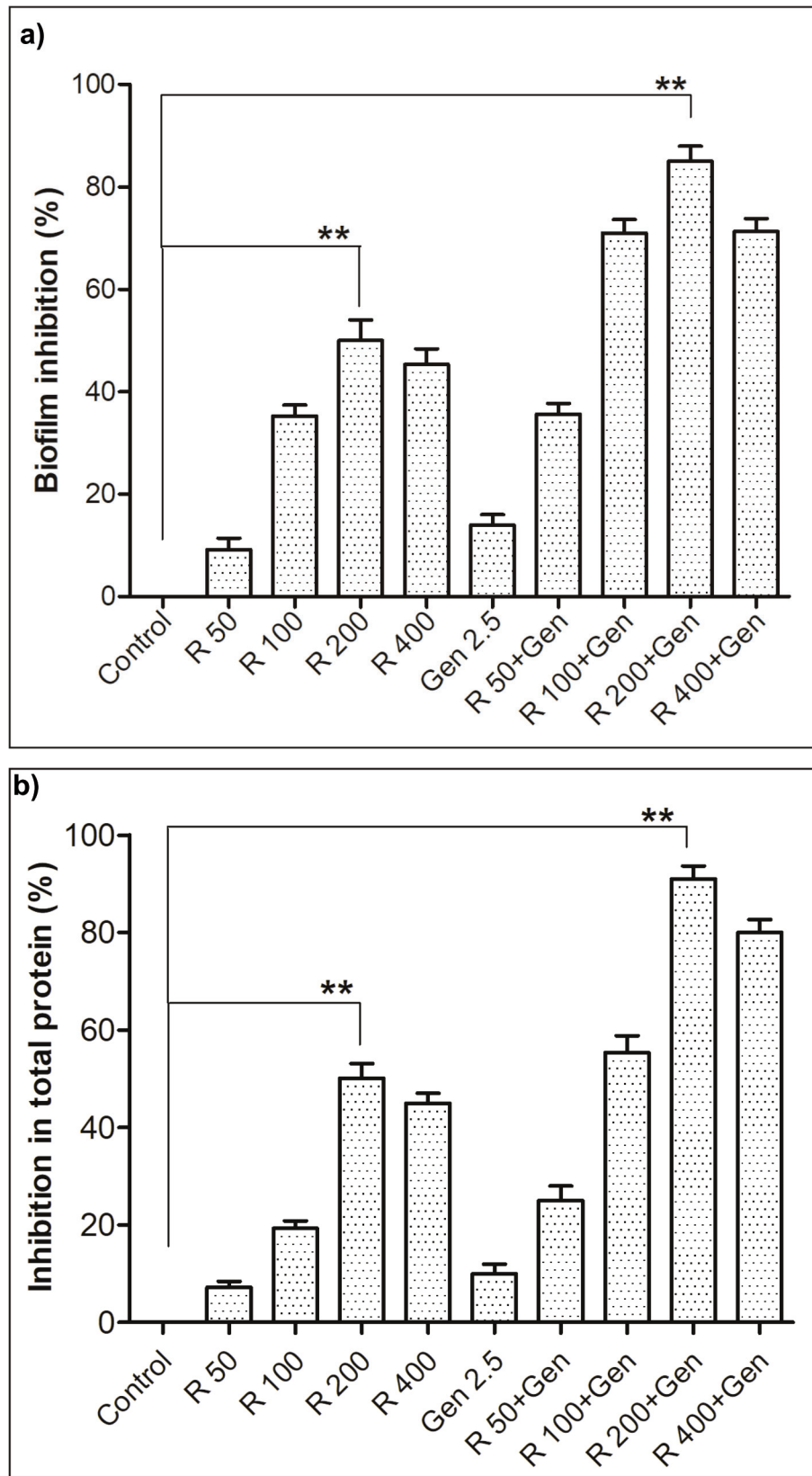
### 3.2.2. Synergistic anti-biofilm activity of rutin in combination with antibiotic

Since rutin exhibits antimicrobial property against *P. aeruginosa*, we hypothesize that rutin may show anti-biofilm activity against the bacterium as well. Anti-biofilm activity of rutin against *P. aeruginosa* was evaluated at several sub-MIC doses (50 µg/ml (1/16 MIC), 100 µg/ml (1/8 MIC), 200 µg/ml (1/4 MIC), 400 µg/ml (1/2 MIC) of the rutin and in combination with sub-MIC dose of gentamicin (2.5 µg/ml (1/4 MIC)) separately. Biofilm inhibition activity of rutin and gentamicin was found to be in a concentration-dependent manner. For rutin alone, the inhibition extent was observed as follows: 9.16% at 50 µg/ml; 35.25% at 100 µg/ml; 50% at 200 µg/ml; and 45.33% at 400 µg/ml whereas it was only 14% for gentamicin treated cells at 2.5 µg/ml (Fig. 4a). Biofilm assay results showed that rutin exerted moderate biofilm attenuation against *P. aeruginosa* where maximum reduction (50%) in biofilm formation was found at 200 µg/ml dose (Fig. 4a). Anti-biofilm activity of rutin increased significantly in combination with the antibiotic gentamicin where maximum synergistic reduction (85%) in biofilm formation was observed at the concentrations of rutin 200 µg/ml with gentamicin 2.5 µg/ml (Fig. 4a). Biofilm total protein assay showed that rutin treated samples had significantly less (maximum inhibition of 50.1% at 200 µg/ml dose) extracted protein in comparison to untreated control (Fig. 4b). Combination of gentamicin with rutin led to significant reduction in total protein content in biofilms. This enhanced extent of reduction in total protein content (91%) in biofilm was observed when 200 µg/ml of rutin was combined with 2.5 µg/ml of gentamicin (Fig. 4b). This observation supported the results of Fig. 4a, wherein the effects of sub-MIC concentrations of Rutin (R) and Gentamicin (Gen) alone and in combination of rutin-gentamicin on the extent of biofilm inhibition in comparison to untreated control was presented. Lou et al. (Lou, Tang, Song, &

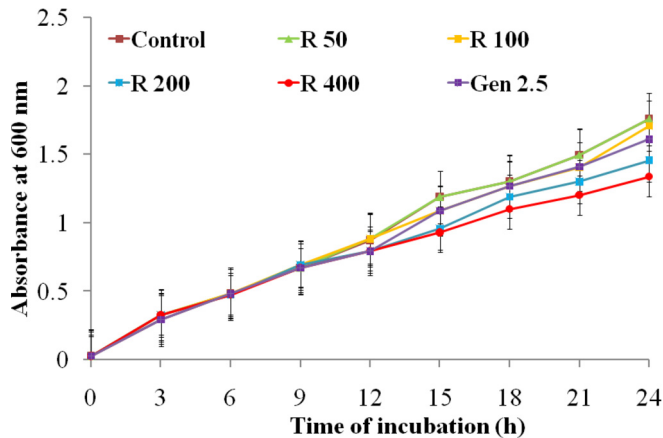
Wang, 2015), reported that rutin was identified as the principal anti-biofilm compound in burdock leaf and it inhibited biofilm formation of *P. aeruginosa* in a metabolomics-based screening study.

The findings of anti-biofilm assay of our study is supported by the earlier report (Gopu, Meena, & Shetty, 2015), that demonstrated substantial attenuation in biofilms of drug resistant microbial pathogens such as *P. aeruginosa*, *Yersinia enterocolitica* and *Klebsiella pneumoniae* by quercetin, a flavonoid, at 80 µg/ml. Another study reported that naringenin and quercetin were able to hinder biofilm formation of *E. coli* and *V. harveyi* (Vikram et al., 2010). Various other polyphenolic compounds of natural origin such as rosmarinic acid, morin, naringin, chlorogenic acid, mangiferin and phloretin have also been reported to inhibit biofilm formed by *P. aeruginosa* (Annapoorani et al., 2012; Lee et al., 2011). Growth curve pattern of *P. aeruginosa* revealed that there was no significant differences were observed in sub-MIC of compound treated *P. aeruginosa* cells (Fig. 5).

Upon extraction and quantification of EPS, it was found that rutin significantly attenuated EPS production of *P. aeruginosa* and the maximum reduction (40.16%) was observed at 200 µg/ml dose (Fig. 6) when compared to untreated control. Further, the degree of reduction was synergistically and significantly enhanced when rutin was combined with gentamicin (83.4%) (Fig. 6). These results clearly indicate that rutin at its sub-MIC level exhibits moderate biofilm attenuation activity which is further increased when it is combined with gentamicin. This synergistic effect of gentamicin and rutin are evidenced by determination of FIC index (Table 2). This study showed flavonoid rutin exhibits moderate antibacterial efficacy against selected bacterial pathogen when combined with conventional antibiotic. Jayaraman et al. (Jayaraman, Sakharkar, Lim, Tang, & Sakharkar, 2010), also reported the antibacterial effect of rutin with various antibiotics against *P. aeruginosa*. EPS production by the microorganisms for biofilm formation is related to the increased resistance of these cells to antibacterial drugs. The



**Fig. 4.** (a). Bar diagrams show the extent of inhibition of biofilm (%) at sub-MIC concentrations of rutin (R), gentamicin (Gen) and combination of both (with respect to control); (b). The extent of total protein reduction in biofilm by *P. aeruginosa* with respect to untreated control. The values are presented as mean  $\pm$  SD based on the assays performed in triplicates. Statistical significance at  $**p \leq 0.005$  ( $n = 3$ ).

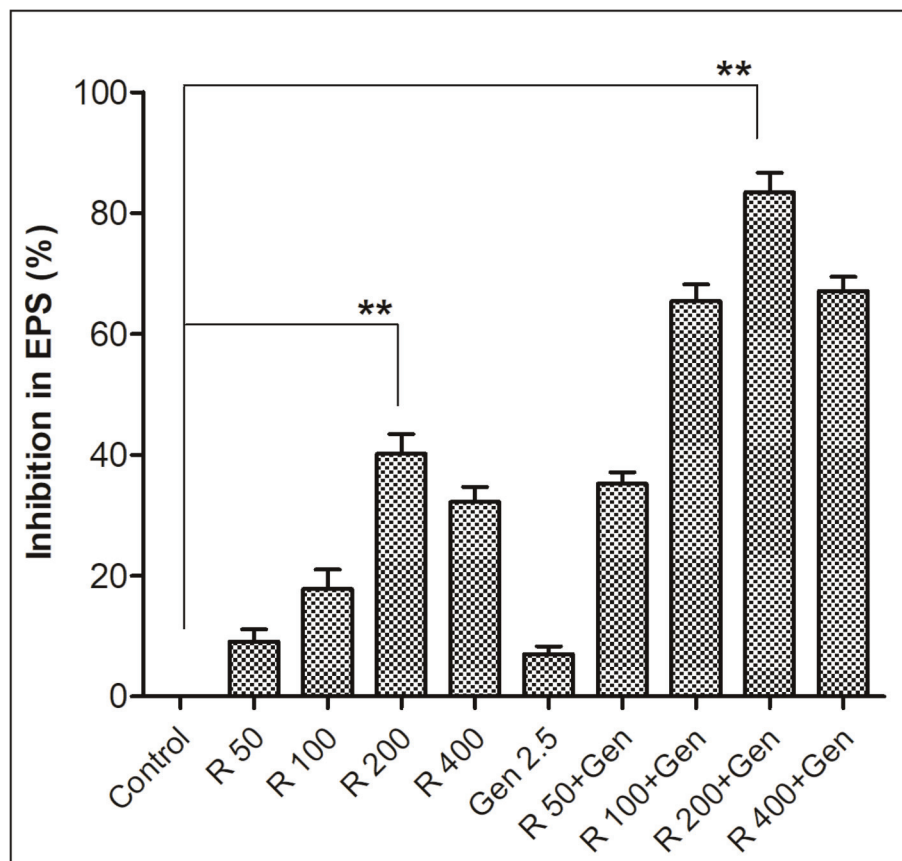


**Fig. 5.** Growth curve of *P. aeruginosa* in LB media in the presence and absence of rutin (R) and gentamicin (Gen) at sub-MIC concentrations. The given data represent the mean value of three independent experiments. Error bars indicate mean  $\pm$  SD.

EPS protects the microbial pathogens in a biofilm by generating multiple layers surrounding the cells in biofilm and it helps the cells

to attach onto the surfaces (Richards & Melander, 2009; Simoes, 2011). Hence, any intrusion with EPS production will apparently lead to the impairment of biofilm formation resulting in the dissociation and exposure of cells to antimicrobial action of drugs as well as body's defense system that threaten the survival of the pathogens. Findings on the inhibition of bacterial biofilm by rutin and rutin-gentamicin correlated well with the attenuated production of EPS after treatment at their sub-MICs (Fig. 6).

The inhibitory activity of rutin, gentamicin and the combination of both were subsequently confirmed with crystal violet staining that showed the attenuation of cell adhesion onto the glass slide surface (Fig. 7). Based on our antibiofilm study using rutin, the concentration of 200  $\mu\text{g/ml}$  was chosen for CLSM analysis. The treated *P. aeruginosa* was stained with acridine orange and observed using Confocal microscope (Fig. 8a). Measurement of biofilm thickness and z-stack imaging (Fig. 8b) revealed a significant difference in biofilm thickness of rutin and gentamicin treated cultures with respect to untreated cultures. As observed in Fig. 8b, thickness of biofilm decreases when treated with rutin and in combination of gentamicin which indicates their widespread inactivation of bacterial growth through its depths. A significant attenuation in attachment of biofilm forming cells was observed at 200  $\mu\text{g/ml}$  concentration of rutin (Fig. 8a) which was supported by

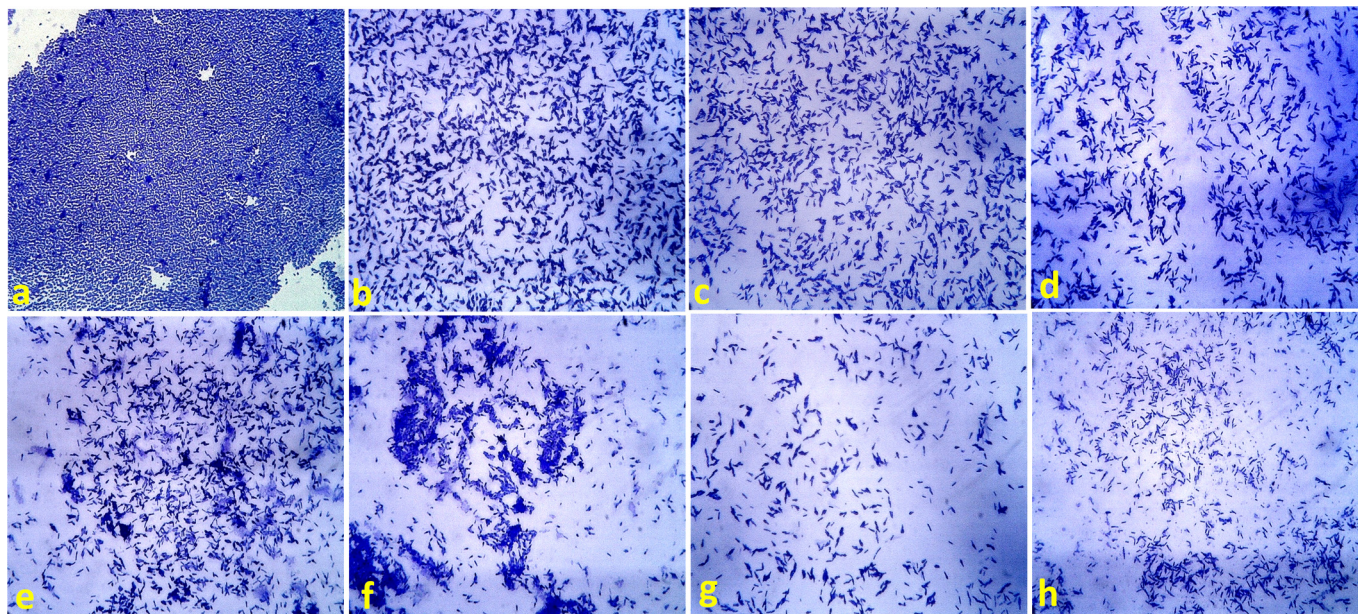


**Fig. 6.** The extent of EPS inhibition (%) from *P. aeruginosa* treated with rutin (R) and gentamicin (Gen) alone, and their combinations with respect to untreated control. The values are presented as mean  $\pm$  SD based on the assays performed in triplicates. Statistical significance at  $**p \leq 0.005$  ( $n = 3$ ).

**Table 2**

Results of interaction study between rutin (R) + gentamicin (Gen). The FIC indexes were interpreted as follows:  $\leq 0.5$  synergy; 0.5–4.0 indifferent and  $> 4.0$  antagonism.

| Strain                         | MIC <sub>Gen</sub> /MIC <sub>R</sub> (alone, $\mu\text{g/ml}$ ) | MIC <sub>Gen</sub> /MIC <sub>R</sub> (in combination, $\mu\text{g/ml}$ ) | FICI (Int) |
|--------------------------------|-----------------------------------------------------------------|--------------------------------------------------------------------------|------------|
| <i>P. aeruginosa</i> MTCC 2488 | 10/800                                                          | 2.5/200                                                                  | 0.50       |



**Fig. 7.** Microscopic images (200 x magnifications) show crystal violet staining of *P. aeruginosa* treated with rutin, gentamicin and combination of both at their sub-MICs. (a) control; (b) rutin (50 µg/ml); (c) rutin (100 µg/ml); (d) rutin (200 µg/ml); (e) rutin (400 µg/ml); (f) gentamicin (2.5 µg/ml); (g) rutin (200 µg/ml + gentamicin 2.5 µg/ml); and (h) rutin (400 µg/ml + gentamicin 2.5 µg/ml).

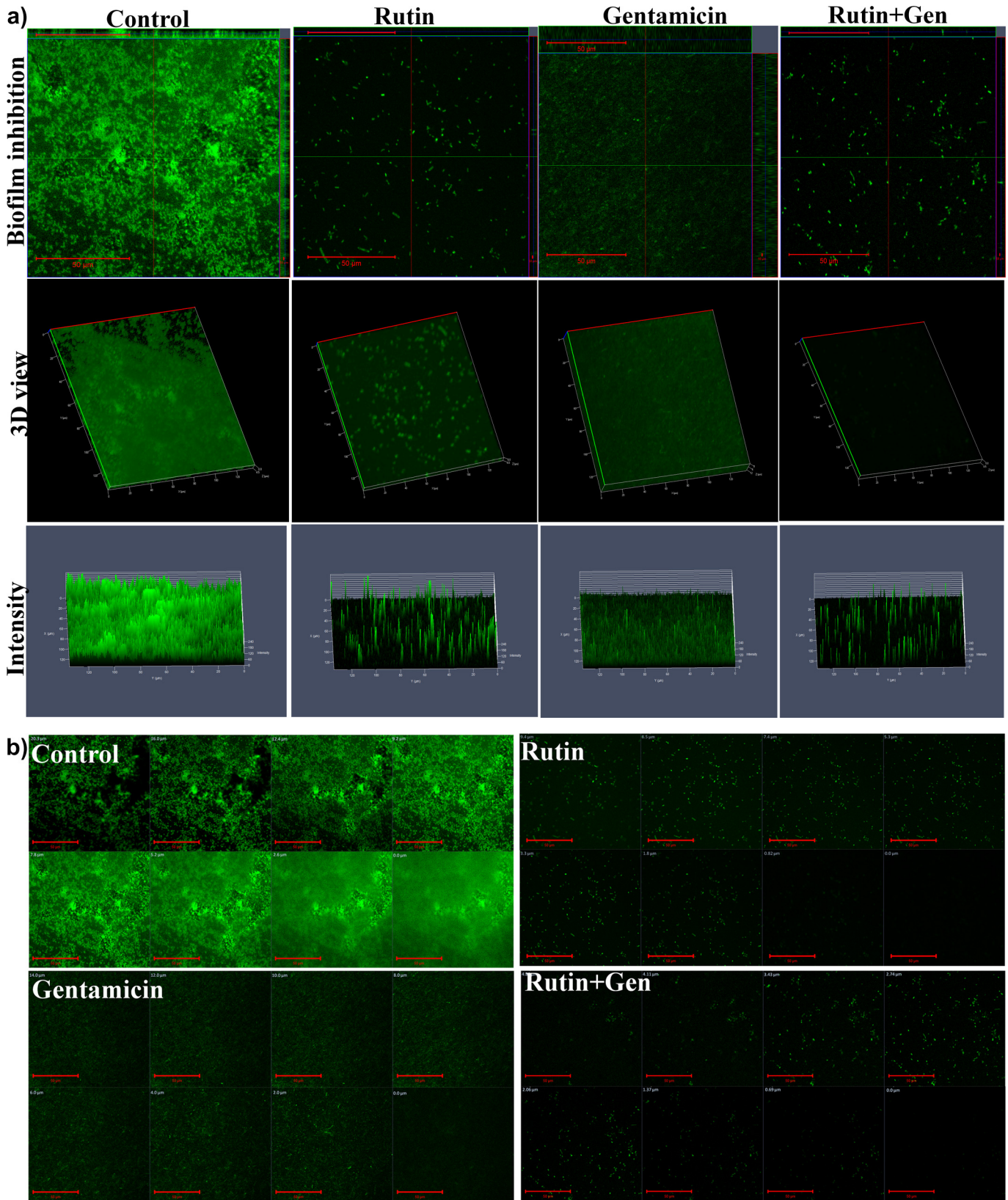
biofilm assay (Fig. 4a), crystal violet staining (Fig. 7) and biofilm total protein extraction assay (Fig. 4b). In untreated control set, very prominent development of biofilm over the glass surface can be seen (Fig. 8a). This clearly indicates that rutin can modulate the formation of biofilm by *P. aeruginosa* on the glass surface and is also evidenced by enhanced thickness reduction in the cells treated with the combination of rutin-gentamicin (Fig. 8a and b). Hence, the results suggest that anti-biofilm potential of rutin increased synergistically in combination with sub-MIC dose of gentamicin.

### 3.2.3. Analysis of endogenous ROS production

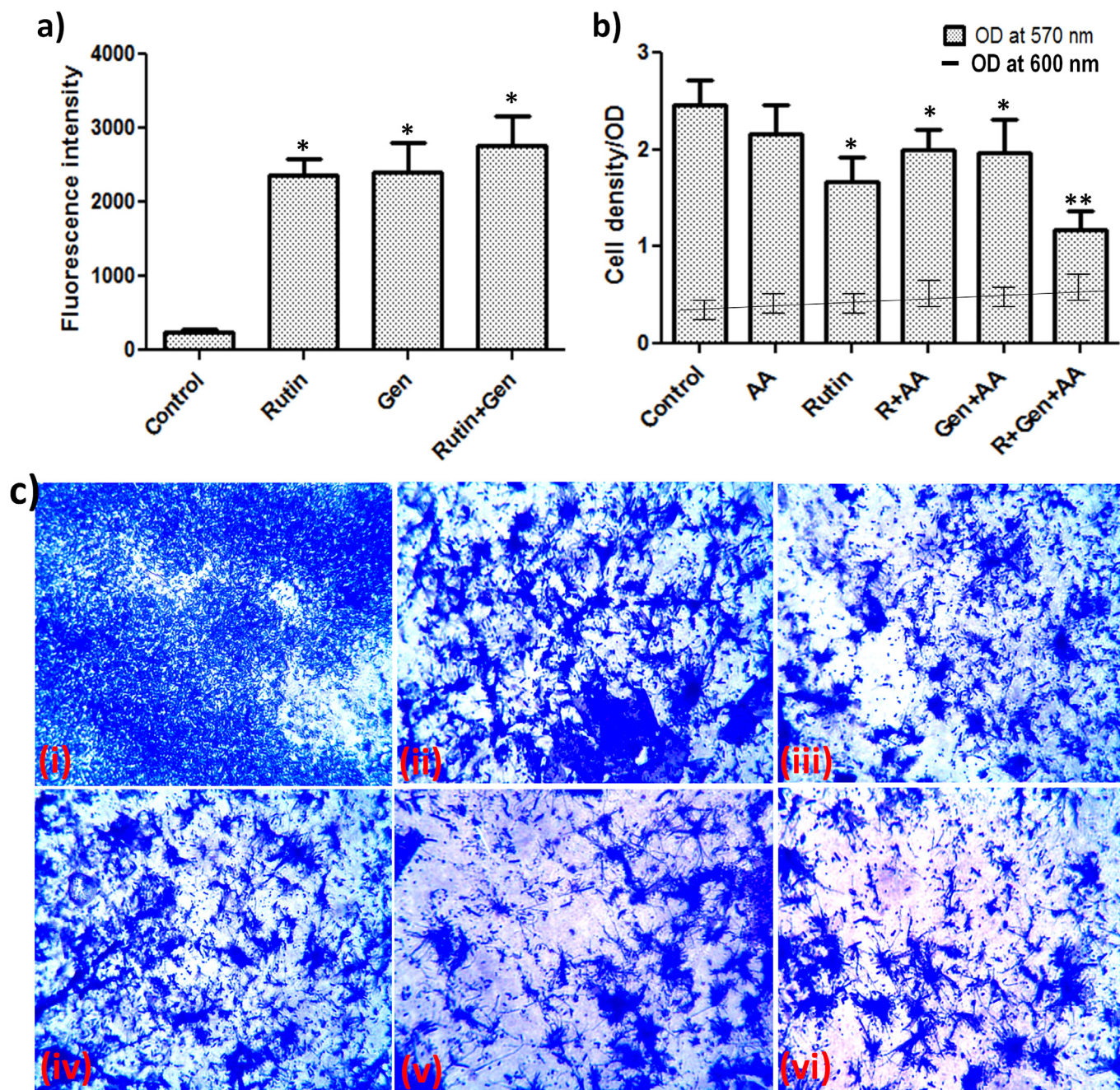
Further, in this study, an attempt was made to evaluate the ability of rutin and its synergistic effect with gentamicin to induce ROS production through DCFDA. The results revealed toxic effects of rutin, gentamicin and combination of rutin-gentamicin through the induction of cell wall disruption, higher membrane permeability and reduced cell density (Fig. 9 (ii)). Generally bactericidal antibiotics such as  $\beta$ -lactam ampicillin, fluoroquinolones ofloxacin and ciprofloxacin, and aminoglycoside gentamicin are shown to cause oxidative stress mediated bacterial cell death due to ROS production (Bhattacharya, Dey, Das, & Banerjee, 2017; Brynildsen, Winkler, Spina, MacDonald, & Collins, 2013; Dwyer et al., 2014). The present study results on the potential of rutin to generate ROS are supported by the findings of previous investigations (Lee, Woo, & Lee, 2016) and the study reported that phytol isolated from *Aster yomena* induced a significant production of ROS in *P. aeruginosa* and exerted toxic effects related to oxidative stress. Several natural polyphenolic compounds such as catechins act on different bacterial species by generating hydrogen peroxide (Haslam et al., 1992) and by altering the permeability of the microbial membrane through ROS generation (Hattori, Kusumoto, Namba, Ishigami, & Hara, 1990). Another study reported that Green tea polyphenols have the ability to act as pro-oxidants to inhibit *P. aeruginosa* and induce the expression of oxidative stress-related genes (Liu et al., 2013). To understand the mechanism of antibacterial activity, ROS generated in the presence of rutin and in combination with gentamicin were studied in the presence of ROS inhibitor AA and

was monitored by oxidation-sensitive fluorescent probe DCFH-DA that passively diffuses through the cell membrane into the cell (Fig. 9a). Citrus flavonoid can affect the cell membrane and lead to ROS formation; DCFH-DA inside the cell reacts with ROS and get converted into fluorescent by-product DCF (Wang et al., 2011). The ROS level was analyzed in the bacteria before and after rutin exposure using cell fluorescence intensity (Fig. 9a). Natural antioxidant such as rutin acts as pro-oxidant that induces the generation of ROS and consequently causes the death of bacteria through damaging the cell membrane, DNA, protein by oxidizing them (Lee et al., 2016; Liu et al., 2013). The anti-biofilm activity of rutin and in combination with gentamicin was quantitatively assessed using crystal violet assay through light microscopic evaluation in the presence of the ROS inhibitor of ascorbic acid (Fig. 9c). The filamentation of *P. aeruginosa* and monolayers of biofilm were found to be maximal in rutin-gentamicin combination (Fig. 9c (iv)) than rutin alone (Fig. 9c (iii)) treated cells, indicating the synergistic effects.

This study substantiates that rutin inhibits biofilm formation by *P. aeruginosa* and this effect is further improved when the phytochemical is combined with antibiotic agent. This study gains significance as the use of rutin can minimize the dose and cost of antibiotics for controlling the *Pseudomonas* infections. Besides, rutin gives scope to treat infections by biofilm forming MDR bacterial strains as the effects of selection pressure of antibiotics are minimized. The attenuation of biofilm development by this phytochemical reduces the virulence and pathogenicity of bacteria due to disruption of their quorum sensing pathways and can be useful in the management of various clinical conditions due to biofilm associated *Pseudomonas* infections such as cystic fibrosis, chronic wounds, sepsis, immune-compromized conditions etc. (Hentzer et al., 2002; Vandeputte et al., 2010). The biofilm inhibition property of rutin improves the host immunological defense against the bacteria as the cell population in biofilm is dissociated resulting in exposure of pathogen to immune system and clearance from the host. Though use of rutin alone is not sufficient to manage the biofilm associated clinical conditions, it can be combined with



**Fig. 8.** (a). CLSM images of *P. aeruginosa* biofilm formation treated with rutin (200  $\mu\text{g/ml}$ ), gentamicin (2.5  $\mu\text{g/ml}$ ) and combination of both. Three dimensional view of biofilm formed by untreated and treated cultures and their 2-dimentional views with their respective intensity projections. Each experiment was performed using three independent cultures and one representative data set is shown here. Scale bars = 50  $\mu\text{m}$ . (b). Z-stack images of the projections indicating the thickness of biofilms for *P. aeruginosa* treated with rutin (R), gentamicin (Gen) and in combination as observed using confocal microscopy. The depth of Z-stack is indicated in left corner of the images using  $\mu\text{m}$ . The observed images revealed the significant differences in biofilm thickness of rutin (9.4  $\mu\text{m}$ ) and in combination with gentamicin (4.80  $\mu\text{m}$ ) treated cultures with respect to their untreated cultures (20.3  $\mu\text{m}$ ). The presented images of Z-stacks are three independent replicate measurements. Scale bars = 50  $\mu\text{m}$ .



**Fig. 9.** (a) Fluorimetric quantification of endogenous ROS production by H2DCF-DA staining of *P. aeruginosa* cells treated with rutin (200 µg/ml), gentamicin (2.5 µg/ml) and in combination at MIC concentrations (\* $p \leq 0.05$ ). Data are expressed as mean  $\pm$  SD; (b) Density of planktonic and biofilm cells in the untreated control, treated groups in the presence and absence of the ROS inhibitor ascorbic acid (AA). Data are expressed as mean  $\pm$  SD. Statistical significance at \* $p \leq 0.05$ ; \*\* $p \leq 0.005$  ( $n = 3$ ); (c) Light microscope images show the formation and disruption of biofilm in control (i) and treatment groups, (ii) AA-50 µg/ml, (iii) rutin-200 µg/ml, (iv) rutin-200 µg/ml + AA-50 µg/ml, (v) gentamicin-2.5 µg/ml + AA-50 µg/ml and (vi) rutin-200 µg/ml + gentamicin-2.5 µg/ml + AA-50 µg/ml in the presence and absence of a ROS inhibitor.

minimal dose of antibiotics to achieve the desired results including the control of emergence of MDR strains. Further, this study envisages the use of natural products in treating the communicable diseases along with the conventional antibiotic therapies.

#### 4. Conclusion

A simple and preparative friendly chromatographic method was optimized for the isolation of bioactive flavonoid rutin from *C. sinensis* peels and the isolated compound was characterized by UV-vis and FT-IR spectroscopic studies. Besides, we have

demonstrated for the first time the anti-biofilm activity of rutin against multidrug-resistant *P. aeruginosa*, the causative agent of nosocomial infections. The flavonoid of this study displayed enhanced anti-biofilm property when combined with conventional antibiotic gentamicin. Even though the mechanism of inhibition of biofilm formation by rutin is unclear, our study indicates the potential use of rutin as a promising anti-biofilm agent against *P. aeruginosa* as it induces ROS generation in bacteria leading to oxidative stress and death of bacteria. Hence, we propose that the natural molecule can be used as nutraceutical or used alone or in combination with other antibiotics to control and treat life

threatening bacterial infections or the associated clinical conditions without any adverse effects. Further studies on the understanding of molecular mechanisms regarding anti-biofilm activity of rutin are in progress.

## Conflict of interest

None.

## Acknowledgements

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